Almost sure and L^p first-order fluctuations of exponential metrics in every dimension

Abstract

Fix an integer $d \geq 2$, and let h be the whole-space log-correlated Gaussian field with unit-sphere average zero. For the common-field exponential metrics, normalized by $\operatorname{med} D_h^\gamma(0, e_1) = 1$, we prove

$$\frac{D_h^\gamma(0, e_1) - 1}{\gamma} \longrightarrow \frac{1}{d} \langle h, \mu_{[0,e_1]} \rangle \qquad \text{almost surely and in every finite } L^p .$$

The almost sure limit holds through all real parameters. The metric characterization and quantitative comparison estimates are those of the supplied research manuscripts; their precise uses are stated below. For $d \geq 3$, the conclusion is conditional on the proposed strong-metric characterization and the supplied higher-dimensional comparison theorem. The L^p proof is common to every dimension: geodesic occupation measures converge to the straight-segment measure in logarithmic energy, and Gaussian differentiation identifies the first-order term. To control the full parameter interval, we obtain a first-order cutoff estimate in dimension two and a sharper initial parameter comparison in higher dimensions. The proof includes the deterministic change from mean-log to point-median normalization.

Contents

1 Introduction **2**

2 Gaussian pairings with segments **4**

3 The common L^p estimate **6**

4 Planar approximation estimates **9**

5 A first-order expansion before the cutoff limit **10**

6 Controlling the full parameter interval **14**

7 Higher-dimensional parameter control **16**

8 Normalization and completion **19**

A Audit record **21**

1 Introduction

Let $d \geq 2$ be fixed. We study exponential metrics of one whole-space log-correlated Gaussian field h on $\mathbb{R}^d$ as their parameter tends to zero. The zeroth-order limit is Euclidean distance. At the unit pair, we prove that the first-order correction converges to the average of h along the straight segment, both almost surely and in every finite L^p. Almost sure convergence is asserted for the full real parameter interval on the original field probability space.

1.1 The field and the metric

Write $B_R = B_R(0)$ for the Euclidean ball, $\overline{B}_R$ for its closure, and $e_1 = (1, 0, \dots, 0)$. Let σ_d be uniform probability measure on S^{d-1}. For $r > 0$ and $x \in \mathbb{R}^d$, write $h_r(x)$ for the Gaussian pairing with uniform probability measure on $\partial B_r(x)$. The field has logarithmic covariance coefficient one and is pinned by $h_1(0) = \langle h, \sigma_d \rangle = 0$. Its covariance kernel is

$$\begin{aligned} G_d(x, y) &= -\log|x - y| + j_d(x) + j_d(y) - c_d, \\ j_d(x) &= \int_{S^{d-1}} \log|x - u| \; \sigma_d(\mathrm{d}u), \qquad c_d = j_d(e_1)\,. \end{aligned} \tag{1.1}$$

The function j_d is continuous and bounded on compact sets. When $d = 2$, this is the whole-plane Gaussian free field, $j_2(x) = \log\max(1, |x|)$, and $c_2 = 0$. For $d > 2$, the field is the log-correlated field specified by (1.1).

For $x \neq y$, define the probability segment measure and its length-weighted version by

$$\mu_{[x,y]} = \int_0^1 \delta_{(1-t)x+ty} \, \mathrm{d}t, \qquad \ell_{[x,y]} = |x - y| \, \mu_{[x,y]}, \qquad \ell_{[x,x]} = 0\,. \tag{1.2}$$

The corresponding Gaussian pairings are

$$H_d = \left\langle h, \mu_{[0,e_1]} \right\rangle, \qquad \mathcal{L}_h(x, y) = \left\langle h, \ell_{[x,y]} \right\rangle, \qquad \mathcal{L}_h(x, x) = 0\,. \tag{1.3}$$

For a continuous function A on $\overline{B}_R \times \overline{B}_R$, write $\|A\|_R = \sup_{x,y \in \overline{B}_R} |A(x, y)|$; the same convention applies with a compact set K in place of $\overline{B}_R$. The pairings in (1.3) are defined by the L^2 limit of smooth pairings. The logarithmic singularity is integrable on a segment, since $\int_0^1 \int_0^1 -\log|s - t| \; \mathrm{d}s \, \mathrm{d}t = 3/2$. Section 2 verifies the continuity and approximation estimates used below.

We use the planar LQG metric when $d = 2$ and the strong exponential metric assignment in the supplied characterization when $d \geq 3$. The metrics are finite length metrics inducing the Euclidean topology. We assume measurability in h, locality, Weyl scaling, and exact similarity covariance. For higher dimensions, the proposed characterization of [1] is an input to this paper. The full-parameter arguments also use the quantitative comparison results of [3, 4]; the exact statements needed are recorded in Sections 4 and 7. Thus the higher-dimensional conclusion is proved under these supplied metric and comparison results.

Until the final normalization, write ρ_h^ξ for the mean-log normalized metric, so that

$$\mathbb{E} \log \rho_h^\xi(0, e_1) = 0, \qquad F_\xi = \log \rho_h^\xi(0, e_1)\,. \tag{1.4}$$

The deterministic normalizing factor is fixed for the original pinned field law and is kept unchanged under field shifts and rescalings. We suppress the subscript h when the field is fixed. For every bounded smooth Cameron–Martin function f, Weyl scaling means that $\rho_{h+f}^\xi = e^{\xi f} \cdot \rho_h^\xi$, where the

right side is the length metric obtained by integrating $e^{\xi f}$ against ρ_h^ξ-arclength. For $r > 0$, exact similarity covariance reads

$$\rho_h^\xi(x + rOu, x + rOv) = r^{\alpha(\xi)} e^{\xi h_r(x)} \rho_{h^{x,r,O}}^\xi(u, v), \qquad h^{x,r,O}(z) = h(x + rOz) - h_r(x)\,. \tag{1.5}$$

Here O is orthogonal, and $h^{x,r,O}$ has the same pinned law as h. No independence between this field and $h_r(x)$ is assumed. We write $\alpha(\xi) = \xi Q_d(\xi)$. The exponent bounds in [5, Proposition 1.1] give

$$1 - \sqrt{2d}\,\xi \le \alpha(\xi) \le 1 + \sqrt{2}\,\xi\,. \tag{1.6}$$

The parameter correspondence and the point-median convention are

$$Q_d(\xi) = \frac{d}{\gamma} + \frac{\gamma}{2}, \qquad \xi = \frac{\gamma}{d_\gamma}, \qquad D_h^\gamma = \frac{\rho_h^{\xi(\gamma)}}{m_{\xi(\gamma)}}, \qquad m_\xi = \operatorname{med} \rho_h^\xi(0, e_1)\,. \tag{1.7}$$

As in [2, (1.4)], d_γ is the metric Hausdorff dimension. We use the lower median $\operatorname{med} X = \inf\{t : \mathbb{P}[X \le t] \ge 1/2\}$. All point and set distances in this paper are unrestricted whole-space distances.

1.2 The first variation

The common-field version is chosen by the simultaneous approximation construction of [3, Theorem 1.1] in dimension two and by the continuous mean-log construction of [4, Section 5] in dimensions at least three. The following theorem concerns these versions.

Theorem 1.1 (First variation in every dimension)**.** *For the metric family and simultaneous version specified above, and for every fixed integer $d \ge 2$,*

$$\frac{D_h^\gamma(0, e_1) - 1}{\gamma} \longrightarrow \frac{1}{d} H_d \qquad \text{almost surely and in } L^p \text{ for every } 1 \le p < \infty\,. \tag{1.8}$$

The almost sure convergence holds through all real $\gamma \downarrow 0$. More precisely, set $\beta_2 = 1/512$ and $\beta_d = 1/24$ for $d \ge 3$. On one event of probability one, for every $0 < \beta < \beta_d$,

$$\left| \frac{D_h^\gamma(0, e_1) - 1}{\gamma} - \frac{1}{d} H_d \right| = O_{h,d,\beta}(\gamma^\beta)\,. \tag{1.9}$$

For every $1 \le p < \infty$ and $0 < r < 1/24$, there are deterministic $C = C(d, p, r)$ and $\gamma_0 > 0$ such that

$$\left\| \frac{D_h^\gamma(0, e_1) - 1}{\gamma} - \frac{1}{d} H_d \right\|_{L^p} \le C\gamma^r, \qquad 0 < \gamma \le \gamma_0\,. \tag{1.10}$$

All dimensions here are fixed; no uniformity in d is claimed. Moment orders and deterministic grid exponents are fixed before the parameter tends to zero, and the permitted small-parameter range may depend on them. The limit is a centered nondegenerate Gaussian. Its variance is

$$\frac{1}{d^2}\left(\frac{3}{2} + 2\int_0^1 j_d(te_1)\,\mathrm{d}t - c_d\right)\,. \tag{1.11}$$

In dimension two this equals $3/8$. The exponents in (1.9) are not optimized.

The planar argument also controls moving endpoints. We retain this consequence because the uniform estimate that proves it is the input to the planar parameter argument.

Theorem 1.2 (Planar compact sets)**.** *When $d = 2$, on one event of probability one, for every compact $K \subset \mathbb{C}$ and every $0 < \beta < 1/512$,*

$$\sup_{x,y\in K} \left| \frac{D_h^\gamma(x,y) - |x-y|}{\gamma} - \frac{1}{2}\mathcal{L}_h(x,y) \right| = O_{h,K,\beta}(\gamma^\beta)\,. \tag{1.12}$$

For each fixed K and $1 \le p < \infty$, the L^p norm of the supremum in (1.12) is at most $C_{K,p}\gamma^{1/512}$ for all sufficiently small γ.

1.3 Proof outline and inputs

At zero parameter the Euclidean minimizer is the straight segment. The field is a distribution, so differentiating its length formally does not justify the first variation. We instead differentiate the logarithmic distance divided by ξ in Gaussian field directions. Its derivative is the probability occupation measure of a constant-speed metric geodesic. Euclidean approximation brings this measure close to the segment measure; a small-ball bound controls their logarithmic energy. Gaussian Poincaré then gives the quantitative L^p estimate in Section 3. This proof uses the uniform metric regularity of [2, Proposition 3.2], and is included here for every dimension.

Fixed-parameter moment estimates leave open excursions between parameter values. The remaining proof controls those excursions in two ways. In the plane, we first establish a spatially uniform first-order expansion before removing the approximation cutoff. Subtracting a smooth coarse field by Weyl reweighting leaves a metric close to a deterministic radial profile. A rigidity argument identifies that profile with Euclidean distance to first order. The cutoff comparison can then start at accuracy smaller than ξ, and finite-cutoff interpolation propagates this accuracy to the full parameter interval. Sections 5 and 6 give this argument.

For $d \ge 3$, the logarithmic increment at distinct deterministic endpoints consists of the similarity term $(\alpha(\xi) - \alpha(\eta)) \log|x-y|$, the common Gaussian term $(\xi - \eta)\left\langle h, \mu_{[x,y]}\right\rangle$, and an $O_{L^p}(a^{1+1/24})$ remainder on $[a, 2a]$. This gives the initial accuracy needed by the increment estimate of [4, Proposition 3.7]. A polynomial-size grid controls the first-order values; the comparison estimate extends the bound to finer grids and then to the full interval. Section 7 carries out this argument with the field pin used by the comparison theorem. Section 8 restores the point median and the γ parameter for both modes of convergence.

The imported regularity and comparison estimates are results in the supplied research manuscripts. Their proofs of metric construction, annular forcing, and effective comparison are not repeated. The new conclusions below use their stated hypotheses and quantitative constants, including the fixed-height bound in the planar cutoff comparison and the initial-comparison clause in higher dimensions.

2 Gaussian pairings with segments

We first control the Gaussian field averaged along segments. The length factor in $\mathcal{L}_h(x,y)$ makes the process continuous even when the endpoints merge. These estimates are used both for the first-order term and for the removal of a smoothed field in the planar argument.

Let $\mathcal{H}_d$ be the Cameron–Martin space of the pinned field, and let $\mathcal{H}_d^*$ denote its dual. The heat kernel is $k_\varepsilon(z) = (\pi\varepsilon^2)^{-d/2}\exp(-|z|^2/\varepsilon^2)$, and $h_\varepsilon = h * k_\varepsilon$. For a signed measure of total mass zero, its squared Cameron–Martin dual norm is its logarithmic energy. Heat convolution contracts this norm:

$$\|\eta * k_\varepsilon\|_{\mathcal{H}_d^*} \le \|\eta\|_{\mathcal{H}_d^*}\,, \qquad \eta(\mathbb{R}^d) = 0. \tag{2.1}$$

Indeed, the Fourier representation is a positive constant times $\int |\widehat{\eta}(\zeta)|^2 \, |\zeta|^{-d} \, \mathrm{d}\zeta$, and $\left|\widehat{k}_\varepsilon\right| \le 1$.

Lemma 2.1 (Weighted segment process)**.** *For each fixed compact ball K, the process $\mathcal{L}_h$ has a continuous version on K^2, equal to zero on the diagonal. Its $1/8$-Hölder seminorm and its supremum have every finite moment. The same bounds hold uniformly for $\mathcal{L}_{h_\varepsilon}$, $0 < \varepsilon \le 1$. Moreover, for every finite p,*

$$\left\| \|\mathcal{L}_{h_\varepsilon} - \mathcal{L}_h\|_K \right\|_{L^p} \le C_{K,p} \varepsilon^{1/8}. \tag{2.2}$$

The supremum of $\mathcal{L}_h$ on K^2 has a Gaussian tail. The versions can be chosen consistently on all compact sets.

Proof. Every unnormalized segment measure satisfies

$$\ell_{[x,y]}(B_s(z)) \le 2s. \tag{2.3}$$

Consider two endpoint pairs at distance at most $u < 1/2$. Their lengths differ by at most $2u$. Couple the smaller common mass using the affine parametrizations; the transport displacement is at most u. The two common-mass measures satisfy (2.3). Truncate the logarithmic kernel below a radius r. The truncated energy of their difference is bounded by $C_K u/r$. Their self-energy truncation errors are bounded by $C_K r$, by (2.3); the mixed singular term has negative coefficient and can be discarded for an upper bound. Choosing $r = \sqrt{u}$ gives squared norm at most $C_K \sqrt{u}$.

The unmatched mass, at most $2u$, can be placed on the longer segment. If its mass is m, that segment has length at least m. Its pinned energy is bounded by $C_K m^2(1 + \log_+(1/m))$. The triangle inequality in $\mathcal{H}_d^*$ therefore gives

$$\mathbb{E} \left| \mathcal{L}_h(x,y) - \mathcal{L}_h(x',y') \right|^2 \le C_K u^{1/2}. \tag{2.4}$$

This also treats the diagonal.

For mollification, couple a segment measure with its convolution by the Gaussian displacement. Its mean displacement is $O(\varepsilon)$. The difference has total mass zero. The same truncated-kernel calculation gives

$$\sup_{x,y \in K} \mathbb{E} \left| \mathcal{L}_{h_\varepsilon}(x,y) - \mathcal{L}_h(x,y) \right|^2 \le C_K \varepsilon^{1/2}.$$

Convolution preserves (2.3). The Gaussian tail of the convolution kernel controls its logarithmic moment outside a compact set. Thus (2.4) holds uniformly for the mollified processes as well.

Gaussian higher moments and the continuity criterion on the $2d$-dimensional endpoint set give the $1/8$-Hölder bound: choose an auxiliary moment order $q > 16d$, increasing it further to obtain any requested moment of the seminorm. For (2.2), use an endpoint grid of mesh ε^2, with $O(\varepsilon^{-4d})$ ordered pairs. The L^q norm of its maximum smoothing error is at most $C_q \varepsilon^{1/4 - 4d/q}$; take $q \ge \max(p, 32d)$. Interpolation has L^p cost $O(\varepsilon^{1/4})$. Finally a continuous Gaussian process on a compact set with bounded expected supremum satisfies the Gaussian supremum inequality, which gives the Gaussian tail. Construct the versions first on integer-radius balls and identify them on overlaps by continuity and a common countable dense set. □

The variable H_d is nondegenerate. Indeed, choose a smooth Cameron–Martin function supported near an interior point of the segment, away from the unit sphere, whose segment integral is nonzero. Its pairing with $\mu_{[0,e_1]}$ is nonzero, so the dual norm of that measure is positive. Integrating (1.1) against the segment measure twice gives (1.11).

3 The common L^p estimate

The argument in this section works in every fixed dimension. We first bound the logarithmic distance by differentiating in the Gaussian field. Similarity covariance then gives a quantitative Euclidean approximation. That approximation forces geodesic occupation measures toward the straight segment, and a second application of Gaussian Poincaré identifies the first-order term. No differentiation in ξ is used.

3.1 Uniform metric regularity

We quote the following metric estimates from [2, Proposition 3.2]. For $R > 0$, write

$$A_\xi(R) = \sup_{x\neq y\in\overline{B}_R} \frac{\rho_\xi(x,y)}{|x-y|^{1/4}}, \quad A_\xi = 1\vee\sup_{n\geq 0} 2^{-2n} A_\xi(2^{n+2}), \quad I_\xi = 1\wedge\inf_{n\geq 0} 2^{-n/4}\rho_\xi(\partial B_{2^{n+2}}, \partial B_{2^{n+3}})\,.$$

For every finite q, uniformly for sufficiently small ξ,

$$\mathbb{E}\left[A_\xi^q + I_\xi^{-q} + \rho_\xi(0,e_1)^{-q}\right] \leq C_{d,q}\,. \tag{3.1}$$

The last term is a special case of the compact separated-distance estimate in the same proposition. Moreover, for each fixed $R, q < \infty$ and $0 < \sigma < 1$,

$$\mathbb{E}\left[\left(\sup_{x\neq y\in\overline{B}_R} \frac{\rho_\xi(x,y)}{|x-y|^\sigma}\right)^q\right] \leq C_{d,R,q,\sigma}\,. \tag{3.2}$$

The allowed upper bound on ξ may depend on the displayed moment orders. These estimates, together with (1.6), are the only quantitative inputs used for the L^p first variation.

The annular lower bound in (3.1) makes every bounded metric ball Euclidean bounded. Since the metric induces the Euclidean topology, it is proper; being a length metric, it is geodesic. If P is a constant-speed geodesic from 0 to e_1, its total length is $e^{F_\xi} \leq A_\xi$. Comparison with the annular distances shows that its image is contained in B_{L_ξ}, where one can take

$$L_\xi = C_d(A_\xi/I_\xi)^4\,. \tag{3.3}$$

This choice works for every such geodesic. Let $\nu_\xi = \int_0^1 \delta_{P(t)}\,\mathrm{d}t$ be its probability occupation measure. Between the first and last visits to a Euclidean ball of radius $s \leq 1$, the path is still minimizing. The bound defining A_ξ therefore gives

$$\nu_\xi(B_s(z)) \leq B_\xi s^{1/4}, \qquad B_\xi = 1 \vee C_d A_\xi (1+L_\xi)^2 e^{-F_\xi}\,. \tag{3.4}$$

Integrating this small-ball bound over logarithmic scales and using the growth of j_d gives

$$\begin{aligned}\|\nu_\xi\|_{\mathcal{H}_d^*}^2 &\leq C_d\,(1 + \log B_\xi + \log(1+L_\xi)) \\ &\leq C_d\left(1 + \log A_\xi + \log I_\xi^{-1} + (F_\xi)_-\right)\,.\end{aligned} \tag{3.5}$$

For example, the positive singular part is bounded by $\int_0^\infty \min\{1, B_\xi e^{-s/4}\}\,\mathrm{d}s \leq 4(1+\log B_\xi)$. Equation (3.1) bounds every fixed moment of the right side of (3.5). It also gives, for every fixed q, a uniform bound on $\mathbb{E}e^{q|F_\xi|}$.

3.2 The logarithmic expansion

Proposition 3.1 (Unit logarithmic distance)**.** *For every fixed $d \geq 2$ and $1 \leq p < \infty$, for all sufficiently small ξ,*

$$\left\| \frac{F_\xi}{\xi} - H_d \right\|_{L^p} \leq C_{d,p} \xi^{1/24} . \tag{3.6}$$

Consequently,

$$\left\| \frac{\rho_\xi(0, e_1) - 1}{\xi} - H_d \right\|_{L^p} \leq C_{d,p} \xi^{1/24} . \tag{3.7}$$

Proof. Choose a countable orthonormal basis (f_j) of $\mathcal{H}_d$ with smooth bounded representatives. Such a basis is obtained from smooth compactly supported functions modulo constants, subtracting their sphere averages. Weyl scaling makes F_ξ Lipschitz along each Gaussian coordinate, with constant $\xi \|f_j\|_\infty$. At a point of coordinate differentiability, an unperturbed minimizing path is a competitor in both directions of the shift. Dividing the two touching inequalities by positive and negative shifts gives

$$\partial_j F_\xi = \xi \int f_j \, \mathrm{d}\nu_\xi, \qquad \|\nabla F_\xi\|_{\mathcal{H}_d} = \xi \, \|\nu_\xi\|_{\mathcal{H}_d^*} . \tag{3.8}$$

The identity holds for every minimizing path at such a field. Fubini and a countable intersection give it simultaneously in all coordinates. The set of constant-speed geodesics has compact nonempty fibers and a Borel graph: the distance identities can be tested at rational times, using continuity of the metric and the paths. A measurable selection is therefore available, and all estimates below are uniform over that choice.

For completeness, the coordinate argument justifies Gaussian Poincaré without presupposing a Sobolev derivative. If W is either F_ξ or $F_\xi/\xi - H_d$, let $W_N = \mathbb{E}[W \mid G_1, \ldots, G_N]$, where G_j are the Gaussian coordinates. The coordinate Lipschitz property gives weak integration by parts, hence $\partial_j W_N = \mathbb{E}[\partial_j W \mid G_1, \ldots, G_N]$ for $j \leq N$. Jensen bounds the finite gradient by the corresponding dual norm in (3.8), or by $\left\|\nu_\xi - \mu_{[0,e_1]}\right\|_{\mathcal{H}_d^*}$ for the residual. These norms have all fixed moments by (3.5). Finite-dimensional Gaussian Poincaré and L^p martingale convergence therefore apply. Since $\mathbb{E}F_\xi = 0$, the first application gives

$$\|F_\xi\|_{L^p} \leq C_{d,p} \xi . \tag{3.9}$$

It suffices to prove these estimates for $p \geq 2$; the smaller orders follow by monotonicity.

We next obtain a quantitative Euclidean approximation. For deterministic endpoints at separation r, similarity covariance gives

$$\rho_\xi(x, y) = r^{\alpha(\xi)} e^X, \qquad X = \xi h_r(x) + F_\xi(h^{x,r,O}), \quad Oe_1 = (y - x)/r .$$

On a fixed endpoint ball, for $0 < r \leq 1$, the spherical-average variance is at most $C + \log(1/r)$. Cauchy–Schwarz, (3.9), and the fixed exponential moments of F_ξ imply

$$\|X\|_{L^{2p}} \leq C_p \xi \sqrt{1 + \log(1/r)}, \qquad \left\|e^X\right\|_{L^{2p}} \leq C_p r^{-C_p \xi^2} .$$

Use $\left|e^X - 1\right| \leq |X| \, (1 + e^X)$ and (1.6). The factors $r^{\alpha(\xi) - C_p \xi^2} \sqrt{1 + \log(1/r)}$ and $r^{1/2} \, |\log r|$ are bounded for sufficiently small ξ. Separations bounded away from zero are easier by the same formulas. Thus

$$\sup_{x,y \in \overline{B}_R} \|\rho_\xi(x, y) - |x - y|\|_{L^p} \leq C_{R,p} \xi .$$

A spatial grid of mesh ξ has $O_R(\xi^{-2d})$ ordered pairs. Take an auxiliary moment $q \geq \max\{p, 4d\}$ and use (3.2) with exponent $1/2$ to fill the grid. This proves

$$\|E_\xi(R)\|_{L^p} \leq C_{R,p}\xi^{1/2}, \qquad E_\xi(R) = \sup_{x,y\in\overline{B}_R} |\rho_\xi(x,y) - |x-y|| \,. \tag{3.10}$$

On the event $E_\xi(10) < c$, a unit geodesic stays in B_{10} and has total length at least $1/2$. Constant metric speed then gives

$$\rho_\xi(0, P(t)) = t\rho_\xi(0, e_1), \qquad \rho_\xi(P(t), e_1) = (1-t)\rho_\xi(0, e_1)\,.$$

The identity $|v - te_1|^2 = (1-t)\,|v|^2 + t\,|v-e_1|^2 - t(1-t)$ implies

$$\sup_{0\leq t\leq 1} |P(t) - te_1| \leq CE_\xi(10)^{1/2} =: \eta_\xi \,. \tag{3.11}$$

On this event the first-and-last-visit argument improves (3.4) to $\nu_\xi(B_s(z)) \leq \widetilde{B}_\xi s^{1/2}$, with every fixed moment of $\widetilde{B}_\xi$ bounded by (3.2).

Put $\lambda = \mu_{[0,e_1]}$. The measures ν_ξ and λ are coupled by $t \mapsto (P(t), te_1)$ with displacement at most η_ξ. Truncate $-\log|x-y|$ at radius s. The truncated kernel is $1/s$-Lipschitz, so the energy of $\nu_\xi - \lambda$ against it is bounded by $C\eta_\xi/s$. The self-energy truncation errors are at most $C(\widetilde{B}_\xi + 1)s^{1/2}$; the mixed error has a negative coefficient. The one-variable pinning terms cancel because the difference has mass zero. Consequently,

$$\|\nu_\xi - \lambda\|^2_{\mathcal{H}^*_d} \leq C\eta_\xi/s + C(\widetilde{B}_\xi + 1)s^{1/2}\,. \tag{3.12}$$

Take $s = \eta_\xi^{2/3}$ after reducing c so that $\eta_\xi < 1$; if $\eta_\xi = 0$, let $s \downarrow 0$. Equations (3.10)–(3.12) and Hölder's inequality give

$$\left\| \|\nu_\xi - \lambda\|_{\mathcal{H}^*_d} \mathbf{1}_{\{E_\xi(10)<c\}} \right\|_{L^p} \leq C_p\xi^{1/24}\,. \tag{3.13}$$

The exponent comes from $\|\nu_\xi - \lambda\| \leq C(1+\widetilde{B}_\xi)^{1/2}E_\xi(10)^{1/12}$ and the $\xi^{1/2}$ error in (3.10). The complementary event has probability $O_N(\xi^N)$ for every fixed N. Its contribution is absorbed by Hölder and the global bound (3.5).

The centered residual $F_\xi/\xi - H_d$ has Gaussian derivative $\nu_\xi - \lambda$. Apply the second Poincaré argument justified above to obtain (3.6). Finally, $|e^z - 1 - z| \leq z^2 e^{|z|}/2$, (3.9), and the fixed exponential moments give

$$\left\| e^{F_\xi} - 1 - F_\xi \right\|_{L^p} \leq \frac{1}{2}\,\|F_\xi\|^2_{L^{4p}} \left\| e^{|F_\xi|} \right\|_{L^{2p}} \leq C_p\xi^2\,. \tag{3.14}$$

Divide by ξ and use (3.6) to obtain (3.7). □

Exact similarity transfers the logarithmic residual to any deterministic segment. Pairing the transformed field in (1.5) gives $\left\langle h, \mu_{[x,y]} \right\rangle = h_r(x) + H_d(h^{x,r,O})$. Hence

$$\log\rho_\xi(x,y) = \alpha(\xi)\log r + \xi H(x,y) + \xi R^{x,y}_\xi, \qquad H(x,y) = \left\langle h, \mu_{[x,y]} \right\rangle, \quad \left\| R^{x,y}_\xi \right\|_{L^p} \leq C_p\xi^{1/24}\,. \tag{3.15}$$

The residual has the same law as $F_\xi/\xi - H_d$, so the constant is independent of the deterministic endpoints. The first-order pairing $H(x,y)$ is the same random variable at every parameter. Its contribution to a parameter increment is therefore $(\xi - \eta)H(x,y)$, which is used in the higher-dimensional argument.

4 Planar approximation estimates

We now record the planar cutoff estimates used to control all small parameters. Let

$$k_\varepsilon(z) = \frac{1}{\pi\varepsilon^2} e^{-|z|^2/\varepsilon^2}, \qquad h_\varepsilon = h * k_\varepsilon, \qquad D^\xi_{\varepsilon,h}(x,y) = \inf_{P:x\to y} \int_P e^{\xi h_\varepsilon(z)} \, |\,\mathrm{d}z|\,,$$

and use mean-log normalization

$$b_\varepsilon(\xi) = \exp\{\mathbb{E}\log D^\xi_{\varepsilon,h}(0,1)\}, \qquad T_{\varepsilon,\xi} = b_\varepsilon(\xi)^{-1} D^\xi_{\varepsilon,h}. \tag{4.1}$$

The mean-log continuum metric is the family ρ_ξ defined in (1.4). In this and the next two sections, $d = 2$ and e_1 is identified with $1 \in \mathbb{C}$.

We use the following statements of [3].

(I1) For each fixed finite moment order, after reducing the upper bound on ξ, unit distances and their reciprocals, compact upper 1/2-Hölder and reverse 2-Hölder constants, and common geodesic escape radii have bounded moments, uniformly in $0 < \varepsilon \le 1$. The compact upper constants have at most polynomial growth in the radius. These statements hold for $T_{\varepsilon,\xi}$ and for ρ_ξ. This is [3, Proposition 3.1].

(I2) For each fixed $R, p < \infty$,

$$\big\| \|T_{\varepsilon,\xi} - |\cdot - \cdot|\|_R \big\|_{L^p} \le C_{R,p}\xi^{1/2}, \qquad 0 < \varepsilon \le 1. \tag{4.2}$$

This follows from [3, Theorem 4.1], by choosing its exponent strictly above 1/2, then using its arbitrary polynomial failure bound and the diameter moments in I1.

(I3) For each fixed ξ, $T_{\varepsilon,\xi} \to \rho_\xi$ in probability locally uniformly as $\varepsilon \downarrow 0$. The simultaneous version of [3, Theorem 1.1] is the almost-sure limit of the full real-cutoff family, uniformly on compact positive parameter intervals and spatial compact sets.

(I4) Put

$$X_t(\xi) = \log(1 + T_{e^{-t},\xi}) - \log(1 + |\cdot - \cdot|). \tag{4.3}$$

For $M \ge N \ge 1$ and a small parameter slab $[a, 2a]$,

$$\mathbb{P}\big[\mathrm{Lip}(X; [a,2a] \times [N,M]) > C_R(M+1)\big] \le C_R e^{-c_R N}. \tag{4.4}$$

The Lipschitz norm takes values in $C(\overline{B}^2_R)$. This is [3, Proposition 5.2].

(I5) There are fixed $\xi_*, M_0, c > 0$ with the following property, for $[a, 2a] \subset (0, \xi_*]$ and $0 < v < 1/4$. Suppose, for a deterministic threshold $S(a,u) \ge 1$, that

$$\sup_{\xi\in[a,2a]} \sup_{t \ge S(a,u)} \mathbb{P}\Big[\big\|T_{e^{-t},\xi} - \rho_\xi\big\|_{16} > u\Big] \le p_{\mathrm{seed}}, \tag{4.5}$$

where $p_{\mathrm{seed}} > 0$ is one fixed sufficiently small constant. Then

$$\mathbb{P}[\|X_t(\xi) - X(\xi)\|_R > v] \le C_R v^{-M_0} \exp(-c v^{M_0} t) \tag{4.6}$$

for each deterministic $\xi \in [a, 2a]$, provided

$$t \ge C_R v^{-M_0}\big[1 + S(a, cv) + \log(C_R/v)\big], \qquad X(\xi) = \log(1 + \rho_\xi) - \log(1 + |\cdot - \cdot|). \tag{4.7}$$

This is [3, Proposition 11.1]. The seed in (4.5) is only a marginal probability bound.

(I6) For every $0 < v < 1/4$, the comparison constants can be chosen so that

$$\mathbb{P}[\|X_t(\xi) - X(\xi)\|_R > v] \le C(a,v)e^{-b(a,v)t} \quad (t \ge N(a,v)), \tag{4.8}$$

where $0 < b \le 1$, and, for fixed integers J_0, B_0,

$$N(a,v) + b(a,v)^{-1} + \log C(a,v) \le \mathsf{T}_{J_0}(C_R a^{-B_0} v^{-B_0}). \tag{4.9}$$

Here $\mathsf{T}_0(x) = x$ and $\mathsf{T}_{j+1}(x) = \exp(\mathsf{T}_j(x))$. This is [3, Theorem 10.1 and (11.13)]. The height J_0 is fixed. These constants use fixed baseline moment orders; increasing moment orders in (4.2) does not change them.

The last two inputs distinguish the new argument from an application of Borel–Cantelli to fixed-parameter estimates. In particular, the fixed height in (4.9) will be used explicitly.

5 A first-order expansion before the cutoff limit

Proposition 5.1 (Uniform first-order marginal)**.** *In dimension two, for every $R, p < \infty$, uniformly for sufficiently small $\xi > 0$ and $0 < \varepsilon \le \xi^{16}$,*

$$\big\| \|T_{\varepsilon,\xi} - |\cdot - \cdot| - \xi \mathcal{L}_h\|_R \big\|_{L^p} \le C_{R,p}\xi^{1+1/512}. \tag{5.1}$$

The corresponding estimate holds for the mean-log continuum metric ρ_ξ, at each deterministic ξ.

5.1 The logarithmic estimate at finite cutoff

Let $F_{\varepsilon,\xi} = \log T_{\varepsilon,\xi}(0,1)$, so that $\mathbb{E}F_{\varepsilon,\xi} = 0$, and let ν be the probability occupation measure of a constant-speed $T_{\varepsilon,\xi}$-geodesic from 0 to 1. Put $\lambda = \mu_{[0,1]}$ and $H_\varepsilon = \langle h, \lambda * k_\varepsilon \rangle$. The geodesic-energy and second Poincaré part of Proposition 3.1 applies at finite cutoff, using Input I2 for the Euclidean approximation. We verify its gradient and global energy bound to make the estimate uniform in ε.

A field shift by a bounded smooth Cameron–Martin function f changes the mollified field by $f * k_\varepsilon$. The touching competitor argument used in (3.8) therefore gives

$$\nabla(F_{\varepsilon,\xi}/\xi - H_\varepsilon) = (\nu - \lambda) * k_\varepsilon \,. \tag{5.2}$$

The coordinate functions remain Lipschitz, since convolution does not increase their supremum norm. The finite-coordinate conditioning and weak integration-by-parts argument in Proposition 3.1 applies verbatim with this derivative; no uniqueness of geodesics is needed.

Let $E_\xi = \|T_{\varepsilon,\xi} - |\cdot - \cdot|\|_{10}$. On $E_\xi < c$, the unit geodesics stay in B_{10}, their total length is at least $1/2$, and the squared-distance identity in (3.11) gives

$$\sup_{0 \le t \le 1} |P(t) - t| \le C E_\xi^{1/2} \,. \tag{5.3}$$

The upper $1/2$-Hölder estimate in Input I1 gives $\nu(B_s(z)) \le Bs^{1/2}$ with uniformly bounded fixed moments of B on this event. The same logarithmic truncation as in (3.12), followed by (4.2), yields

$$\Big\| \|\nu - \lambda\|_{\mathcal{H}_2^*} \mathbf{1}_{\{E_\xi < c\}} \Big\|_{L^p} \le C_p \xi^{1/24} \,.$$

To control the exceptional event, let L_{10} be a common escape radius from Input I1. Let $H_j \geq 1$ be the least upper 1/2-Hölder constant for separations at most one in B_{2^j}, and define $H_* = 1 + \sup_{j\geq 1} e^{-j^2} H_j$. For every fixed q, the polynomial-radius moment estimate gives

$$\mathbb{E}H_*^q \leq C_q \left(1 + \sum_{j\geq 1} e^{-qj^2} 2^{A_q j}\right) < \infty\,.$$

The weights are fixed independently of q; the polynomial radius exponent A_q may depend on q. On the escape ball, the spatial constant is at most $H_* \exp\{C[1+\log(1+L_{10})]^2\}$. The first-and-last-visit estimate and logarithmic integration therefore bound the occupation energy by

$$C\left[1 + \log^2(1+L_{10}) + \log H_* + \log_+(T_{\varepsilon,\xi}(0,1)^{-1})\right]\,.$$

This has every fixed moment uniformly in the cutoff. Since $\mathbb{P}[E_\xi \geq c] = O_N(\xi^N)$ for every fixed N, Hölder's inequality absorbs the exceptional event. Heat convolution contracts the dual norm of the zero-mass difference by (2.1). Gaussian Poincaré, applied to the centered residual in (5.2), now gives

$$\|F_{\varepsilon,\xi}/\xi - H_\varepsilon\|_{L^p} \leq C_p \xi^{1/24}, \qquad 0 < \varepsilon \leq 1\,. \tag{5.4}$$

In particular, $\|F_{\varepsilon,\xi}\|_{L^p} \leq C_p \xi$. Fixed exponential moments follow from Input I1.

5.2 A deterministic radial profile

Rescaling a finite-cutoff distance changes the cutoff. The centered Gaussian estimate therefore leaves a deterministic dependence on the endpoint separation to identify. We record that dependence in the expected logarithm by defining

$$g_{\varepsilon,\xi}(r) = \exp\{\mathbb{E}\log T_{\varepsilon,\xi}(0,r)\} \quad (r>0), \qquad g_{\varepsilon,\xi}(0) = 0. \tag{5.5}$$

Then $g_{\varepsilon,\xi}(1) = 1$. Translation and rotation of the pinned field change a log distance by a centered Gaussian constant. Consequently $\mathbb{E}\log T_{\varepsilon,\xi}(x,y) = \log g_{\varepsilon,\xi}(|x-y|)$ for every deterministic pair. The profile is continuous away from zero by local regularity and uniform integrability of separated log distances. At zero, Jensen and the upper spatial moment bound give $g_{\varepsilon,\xi}(r) \leq Cr^{1/2}$, so it is continuous there too.

For $r = |x-y| \geq \xi^8$ and $\varepsilon \leq \xi^{16}$, exact heat-LFPP scaling and (5.4) give

$$\log T_{\varepsilon,\xi}(x,y) = \log g_{\varepsilon,\xi}(r) + \xi H_\varepsilon(x,y) + \xi R^{x,y}_{\varepsilon,\xi}, \qquad \left\|R^{x,y}_{\varepsilon,\xi}\right\|_{L^p} \leq C_p \xi^{1/24}, \tag{5.6}$$

where $H_\varepsilon(x,y) = \left\langle h_\varepsilon, \mu_{[x,y]}\right\rangle$. Indeed, after rescaling by r, the cutoff is $\varepsilon/r \leq 1$, and the field is $h(x + rO\cdot) - h_r(x)$, with its original pinned law. The identity for H_ε retains the same circle average, so the first-order term in (5.6) is the pairing with the original field. No independence is used.

Write $Z = \log T_{\varepsilon,\xi}(x,y) - \log g_{\varepsilon,\xi}(r)$. It is centered, and, for $r \geq \xi^8$ in a fixed endpoint ball,

$$\|Z\|_{L^p} \leq C_p \xi \sqrt{1 + \log_+(1/r)}, \qquad \mathbb{E}e^{q|Z|} \leq C_q.$$

The second assertion follows from the Gaussian circle-average moment and the unit logarithm moment by Cauchy–Schwarz. Taylor's inequality $|e^z - 1 - z| \leq z^2 e^{|z|}/2$ now gives

$$\|T_{\varepsilon,\xi}(x,y) - g_{\varepsilon,\xi}(r) - \xi g_{\varepsilon,\xi}(r) H_\varepsilon(x,y)\|_{L^p} \leq C_p \xi^{1+1/48}, \tag{5.7}$$

$$|\mathbb{E}T_{\varepsilon,\xi}(x,y) - g_{\varepsilon,\xi}(r)| \leq C\xi^2 \log(1/\xi). \tag{5.8}$$

The factor $g_{\varepsilon,\xi}$ is bounded on compact intervals by Jensen and the distance moments. Together with (4.2), this proves

$$\sup_{0\le r\le 2R} |g_{\varepsilon,\xi}(r) - r| \le C_R \xi^{1/2}. \tag{5.9}$$

For $r < \xi^8$, use $g(r) \le Cr^{1/2}$ directly. We may therefore replace $g(r)$ by r in the linear term of (5.7).

Use an endpoint grid of mesh ξ^{32} in a fixed ball. It has $O_R(\xi^{-64d})$ ordered pairs, with $d = 2$ here. Choose a fixed auxiliary moment $P \ge \max(p, 12288d)$. The L^p norm of the grid maximum costs at most $\xi^{-64d/P} \le \xi^{-1/192}$. The metric interpolation cost is $O_{L^p}(\xi^{16})$. For the profile, compare to $\mathbb{E}T_{\varepsilon,\xi}(0,r)$ by (5.8); its interpolation cost is $C\xi^{16} + C\xi^2 \log(1/\xi)$. Lemma 2.1 controls the linear Gaussian term. Pairs with separation at most $2\xi^8$ have metric and profile bounded by an all-moment factor times ξ^4, and their $\xi\mathcal{L}_h$ term is $O_{L^p}(\xi^2)$. Finally (2.2) costs at most $\xi\varepsilon^{1/8} \le \xi^3$. We conclude that

$$\left\| \sup_{x,y\in \overline{B}_R} |T_{\varepsilon,\xi}(x,y) - g_{\varepsilon,\xi}(|x-y|) - \xi\mathcal{L}_h(x,y)| \right\|_{L^p} \le C_{R,p}\xi^{1+1/192}. \tag{5.10}$$

5.3 Removing a smooth coarse field

Fix the endpoint ball B_R and set $M_1 = \lceil \max\{1, 2R\} \rceil$, $R_1 = 20(M_1 + 1)$, and $R_2 = 20R_1$. Choose a smooth compactly supported function χ equal to one on B_{R_2}. These choices are deterministic and depend only on R. The radial approximation will be proved on B_{R_1}, which will also contain the geodesics needed for rigidity. Set $\delta = \xi^{1/32}$ and put

$$f = \chi h_\delta, \qquad M = \|f\|_\infty, \qquad L = \|\nabla f\|_\infty, \qquad \overline{T} = e^{-\xi f} \cdot T_{\varepsilon,\xi}.$$

This is a pathwise Weyl reweighting of a smooth length metric; it makes no assertion about the law of a randomly shifted field. Since f is bounded, $\overline{T}$ is proper and geodesic, and

$$e^{-\xi M} T_{\varepsilon,\xi} \le \overline{T} \le e^{\xi M} T_{\varepsilon,\xi}. \tag{5.11}$$

On each fixed support,

$$\|M + L\|_{L^p} \le C_p \delta^{-2}, \qquad \mathbb{E}e^{q\xi M} \le C_q. \tag{5.12}$$

For the exponential bound, use the actual Gaussian supremum estimate: the expected supremum is $O(\log(1/\delta))$ and its maximal variance is $O(\log(1/\delta))$. The crude power bound on M alone would not justify it.

Let $E = \|T_{\varepsilon,\xi} - |\cdot - \cdot|\|_{R_2}$. For a sufficiently small fixed $c = c_R > 0$, consider the event $E < c$ and $\xi M < c$. On this event, every geodesic for either metric with endpoints in B_{R_1} stays in B_{R_2}. Indeed, its full distance is at most $e^c(2R_1 + c)$, whereas reaching ∂B_{R_2} from B_{R_1} costs at least $e^{-c}(R_2 - R_1 - c)$, by (5.11). The latter is larger. The same comparison shows separately that the geodesics from 0 to e_1 and from 0 to $M_1 e_1$ stay in B_{R_1}: their total distance is at most $e^c(M_1 + c)$, whereas reaching ∂B_{R_1} costs at least $e^{-c}(R_1 - c)$. On B_{R_2}, the Euclidean error of $\overline{T}$ is at most $C_R(E + \xi M)$.

Choose the preceding good-event constant c_R small enough that the Euclidean errors of both metrics on B_{R_2} are at most one. The geodesics just considered lie in that ball. If the corresponding metric error is $e \le 1$, the squared-distance identity used in (5.3) puts each constant-speed geodesic within $C_R\sqrt{e}$ of $(1-t)x + ty$. Writing D for either metric along its geodesic, it follows that

$$\left| \int f \, \mathrm{d}\,\mathrm{len}_D - |x-y| \left\langle h_\delta, \mu_{[x,y]} \right\rangle \right| \le C_R(Me + L\sqrt{e}).$$

Expand $e^{-\xi f}$ along a $T_{\varepsilon,\xi}$-geodesic for the upper comparison. Expand $e^{\xi f}$ along an $\overline{T}$-geodesic for the reverse comparison. The two estimates give

$$\left\|\overline{T} - T_{\varepsilon,\xi} + \xi\mathcal{L}_{h_\delta}\right\|_{R_1} \le C_R\{\xi M(E+\xi M) + \xi L\sqrt{E+\xi M} + \xi^2 M^2 e^{C\xi M}\}. \tag{5.13}$$

The norms in (4.2) and (5.12) make its L^p norm $O(\xi^{1+3/16})$. For example, the largest term is bounded by $\xi\, O_{L^{2p}}(\xi^{-1/16}) O_{L^{2p}}(\xi^{1/4})$. The exceptional event has probability $O_N(\xi^N)$ for every fixed N, by high fixed moments. Hölder's inequality, (5.11), and (5.12) absorb its contribution to every requested L^p norm.

Applying (5.10) on the fixed ball B_{R_1} and combining it with (5.13) and Lemma 2.1 at scale δ, we obtain

$$\left\| \sup_{x,y\in\overline{B}_{R_1}} \left|\overline{T}(x,y) - g_{\varepsilon,\xi}(|x-y|)\right| \right\|_{L^p} \le C_{R,p}\xi^{1+1/256}. \tag{5.14}$$

The slowest error here is $\xi\delta^{1/8} = \xi^{1+1/256}$.

5.4 Radial rigidity

Lemma 5.2 (Deterministic radial rigidity)**.** *Let $M_1 \ge 1$ be an integer. Suppose a continuous geodesic metric D and a continuous nonnegative function g, with $g(0) = 0$ and $g(1) = 1$, satisfy $|D(x,y) - g(|x-y|)| \le \eta$ for some $0 < \eta < 1/4$ on a ball containing $[0, M_1 e_1]$ and geodesics from 0 to e_1 and from 0 to $M_1 e_1$. Then*

$$\sup_{0\le r\le M_1} |g(r) - r| \le C_{M_1}\eta(1 + \log(1/\eta)). \tag{5.15}$$

Proof. Partition a minimizing path by successive first exits from Euclidean balls of radius r about its current point. There are finitely many full steps, by uniform continuity of the parametrized path, and one residual step shorter than r. For endpoint separation L_0, the number N of full steps satisfies $N \ge \lfloor L_0/r \rfloor$. On the unit geodesic, take $r = 1/n$. Additivity of geodesic subpath lengths and straight concatenation give

$$g(1/n) \le \eta + (1+\eta)/n, \qquad 1 - \eta \le n(g(1/n) + \eta).$$

Thus $|g(1/n) - 1/n| \le (1 + 1/n)\eta$. The same exit argument gives, for $0 < t \le 1/2$,

$$g(t) \le \eta + \frac{(1+\eta)t}{1-t} \le t + 2t^2 + 2\eta.$$

Write $r \in [0, M_1]$ as its integer part, its first K binary digits, and a remainder below 2^{-K}. Concatenate these straight increments and take $K = \lceil \log_2(1/\eta) \rceil$. This proves the upper bound in (5.15). The 0-to-$M_1 e_1$ geodesic has at least M_1 full unit steps, so $g(M_1) \ge M_1 - (M_1 + 1)\eta$. The triangle inequality through re_1, together with the upper bound for $g(M_1 - r)$, proves the lower bound. □

Completion of Proposition 5.1. Use the fixed radii R_1, R_2 and the same reweighted metric constructed above. The event $E < c$, $\xi M < c$ has probability at least $3/4$ for all sufficiently small ξ, uniformly over the allowed cutoffs. On it, the two geodesics required by Lemma 5.2 stay in B_{R_1}. By (5.14) and Markov's inequality, the radial approximation on that same ball has error at most

$A\xi^{1+1/256}$ with probability at least 3/4, for one fixed A. The intersection has positive probability. Lemma 5.2 applies there on B_{R_1}, and its conclusion concerns only the deterministic function $g_{\varepsilon,\xi}$. Hence, using $2R \le M_1$, it holds deterministically:

$$\sup_{r\le 2R} |g_{\varepsilon,\xi}(r) - r| \le C_R\xi^{1+1/256}(1+\log(1/\xi)) \le C_R\xi^{1+1/512}.$$

Insert this into (5.10). For the continuum statement, fix ξ, take an almost-sure subsequence of the convergence in Input I3, and apply Fatou. This uses only a countable cutoff subsequence at this one deterministic parameter. □

6 Controlling the full parameter interval

The cutoff expansion provides an initial error smaller than ξ. We combine it with the effective comparison from Section 4 to control the continuum family for all small parameters. Interpolation is performed only on finite-cutoff metrics, whose parameter dependence is quantified.

Proposition 6.1 (Planar parameter estimate)**.** *On one event of probability one, for every $R < \infty$ and every $0 < \beta < 1/512$,*

$$\|\rho_\xi - |\cdot - \cdot| - \xi\mathcal{L}_h\|_R = O_{h,R,\beta}(\xi^{1+\beta}) \qquad (\xi \downarrow 0)\,.$$

The limit is through all real parameters.

Proof. Set $\nu_0 = 1/512$. Proposition 5.1 is the additional estimate used in this proof. The parameter sets on which probability bounds are first applied are finite deterministic grids.

The initial comparison accuracy. Fix $0 < \beta < \kappa < \nu_0$, let a be small, and put

$$e = a^{1+\kappa}, \qquad u = a^{1+\beta}.$$

For $\xi \in [a, 2a]$ and $t \ge 16\log(1/a)$, both the cutoff and continuum distances have the same first-order approximation $|x-y| + \xi\mathcal{L}_h(x,y)$. Hence

$$\Big\| \|T_{e^{-t},\xi} - \rho_\xi\|_{16} \Big\|_{L^p} \le C_p a^{1+\nu_0}.$$

For any fixed multiple of e, the seed (4.5) therefore holds once a is sufficiently small, with

$$S(a, ce) \le 16\log(1/a) + C. \tag{6.1}$$

Only one fixed moment order is needed to obtain the prescribed seed confidence. Applying Input I5, the comparison at accuracy e has constants N_F, b_F, C_F satisfying, for a fixed finite B_F,

$$b_F \ge ca^{B_F}, \qquad N_F + \log C_F \le Ca^{-B_F}. \tag{6.2}$$

The exponent B_F is fixed before any large moment used to control a parameter grid.

Define the random continuous profile

$$V_h(x,y) = \frac{\mathcal{L}_h(x,y)}{1+|x-y|}, \qquad Y_t(\xi) = X_t(\xi) - \xi V_h.$$

Taylor expansion of $\log(1+s)$, whose second derivative is bounded on $s \ge 0$, and Proposition 5.1 give

$$\|\|Y_t(\xi)\|_R\|_{L^p} \le C_{R,p}\xi^{1+\nu_0}, \qquad t \ge 16\log(1/\xi). \tag{6.3}$$

The Gaussian tail of $\|V_h\|_R$ lets us retain the finite-cutoff Lipschitz bound (4.4) for Y, after increasing constants. Also $Y_t(\xi) - Y_s(\xi) = X_t(\xi) - X_s(\xi)$.

One cutoff transition. Suppose (4.8) holds at tolerance v. On the Lipschitz event, grid $[a,2a]\times[N,M]$ at mesh $v/[8C_R(M+1)]$. Its cardinality is at most

$$C_R[1+a(M+1)/v]\,[1+(M+1)^2/v].$$

Compare $X_t(\xi)$ and $X_N(\xi)$ at each grid point to the same fixed-parameter variable $X(\xi)$. Interpolation, including interpolation of the anchor, gives

$$\mathbb{P}\left[\sup_{\substack{\xi\in[a,2a]\\ N\le t\le M}}\|X_t(\xi)-X_N(\xi)\|_R>3v\right]\le C_R[1+a(M+1)/v][1+(M+1)^2/v]\cdot C(a,v)e^{-b(a,v)N}+C_Re^{-c_RN}. \tag{6.4}$$

If

$$N\ge C_R\left\{N(a,v)+b(a,v)^{-1}[1+\log C(a,v)+\log(1/v)+\log(1/b(a,v))]\right\}, \tag{6.5}$$

we may take

$$M=\lfloor\exp\{b(a,v)N/64\}\rfloor\ge 2N. \tag{6.6}$$

The right-hand side of (6.4) is then at most $C_Re^{-c_Rb(a,v)N}$. The grid costs only a fixed power three of M. This statement controls every real cutoff between N and M.

Reaching arbitrary cutoff depths. For a summable accuracy sequence $v_k=e2^{-k-1}$, the tower bound (4.9) implies that, for fixed sufficiently large A,B_1,

$$z_k=Aa^{-B_1}2^{B_1k},\qquad A_k=\mathsf{T}_{J_0+2}(z_k)$$

dominate the thresholds in (6.5). Increasing A,B_1 if necessary, for every $m_k\ge A_k$,

$$b(a,v_k)m_k\ge\sqrt{m_k},\qquad \left\lfloor e^{\sqrt{A_k}/64}\right\rfloor\ge A_{k+1}. \tag{6.7}$$

For the first inequality, A_k exceeds $\mathsf{T}_{J_0}(z_k)^2$; the second follows by comparing successive logarithms of the two fixed-height towers. Thus a start $m_0\ge A_0$ permits infinitely many transitions with total error at most $3e$ and total failure at most $C_Re^{-c_R\sqrt{m_0}}$.

The remaining issue is reaching A_0 from a polynomial depth without paying a tower-sized parameter grid. Let $J=J_0+4$. Choose a fixed

$$L_0>2B_F+B_1+10,\qquad n_0=\lceil a^{-L_0}\rceil.$$

Increase L_0 further if a fixed numerical margin requires it. At time n_0, a parameter grid of mesh $u/[4C_R(n_0+1)]$ has $O(a^{-L_0-\beta})$ points. For any prescribed failure power $m>0$, choose one fixed moment P with

$$P(\nu_0-\beta)>m+L_0+\beta+2.$$

Equation (6.3), Markov's inequality, and the Lipschitz event imply

$$\mathbb{P}\left[\sup_{\xi\in[a,2a]}\|Y_{n_0}(\xi)\|_R>2u\right]\le C_ma^m. \tag{6.8}$$

This high moment is chosen after B_F,B_1,L_0. Its only effect on previous estimates is a possible reduction of the allowed upper parameter.

Make J transitions at the fixed accuracy e, using (6.2):

$$n_{j+1} = \left\lfloor e^{b_F n_j/64} \right\rfloor, \qquad 0 \le j < J.$$

Their combined failure is at most $C_J e^{-ca^{B_F} n_0} = O(a^m)$. Their accumulated error is $3Je = o(u)$. They reach $n_J \ge A_0$ for all sufficiently small a. Here is a direct tower comparison. Put $x = Aa^{-B_1}$. The choice of L_0 gives $b_F n_0/64 \ge (J+3)x$. Then $n_1 \ge \mathsf{T}_1((J+2)x)$. At each of the finitely many next steps, multiplication by $b_F/64$, a reciprocal polynomial in a, is absorbed by decreasing the top input by x:

$$(b_F/64)\mathsf{T}_j(c_1 x) \ge \mathsf{T}_j((c_1-1)x), \quad j \ge 1,\ c_1 \ge 2.$$

The exponential ratio already proves this at $j = 1$; higher levels are larger. Fixed margins absorb integer rounding. Hence $n_J \ge \mathsf{T}_J(2x) \ge A_0$.

Now use the summable transitions (6.7). On their complete success the total additional error is at most $3e$, and every real later cutoff is controlled. The full-cutoff limit from Input I3 consequently satisfies

$$\mathbb{P}\left[\sup_{\xi\in[a,2a]} \|X(\xi) - \xi V_h\|_R > C_R a^{1+\beta}\right] \le C_{R,m} a^m. \tag{6.9}$$

Only countably many finite grids and cutoff intervals were used.

Apply (6.9) to $a = 2^{-j}$, then Borel–Cantelli, integer spatial balls, and a countable sequence of exponents increasing to ν_0. Exponentiating on each compact set gives, on one event of probability one,

$$\|\rho_\xi - |\cdot - \cdot| - \xi\mathcal{L}_h\|_R = O_{h,R,\beta}(\xi^{1+\beta}), \qquad 0 < \beta < 1/512, \tag{6.10}$$

through all real $\xi \downarrow 0$. □

7 Higher-dimensional parameter control

Fix $d \ge 3$. We use the common L^p expansion to compare nearby parameters on the same field. The common pairing contributes $(\xi - \eta)H(x, y)$, while the residuals contribute $O_{L^p}(a^{1+1/24})$ on a slab $[a, 2a]$. This gives a smaller error than the two separate Euclidean approximations. The increment estimate of [4, Proposition 3.7] then controls the gaps between a polynomial number of initial parameter values.

Proposition 7.1 (Higher-dimensional parameter estimate)**.** *For the simultaneous mean-log metric with unit-sphere pinning, on one event of probability one, for every* $0 < \beta < 1/24$*,*

$$\rho_\xi(0, e_1) = 1 + \xi H_d + O_{h,d,\beta}(\xi^{1+\beta}) \qquad (\xi \downarrow 0)\,.$$

The limit is through all real parameters.

7.1 The comparison input and a finer initial seed

Write ρ_ξ for mean-log normalization and $\alpha(\xi) = \xi Q(\xi)$. The function α is Lipschitz near zero [5, proof of Lemma 3.7]; this is also recorded with a finite-cutoff proof in [4, Lemma 2.1].

The comparison in [4, Sections 2–5] uses the field pinned on the sphere of radius ten. We therefore carry out the entire bootstrap with that pin. More precisely, let $h^{(1)}$ denote the field in Proposition 7.1, and put

$$U = h^{(1)}_{10}(0), \qquad h^{(10)} = h^{(1)} - U, \qquad \rho^{(10)}_\xi = e^{-\xi U}\rho^{(1)}_\xi.$$

Both families are mean-log normalized, since $\mathbb{E}U = 0$. The Weyl identity at each deterministic parameter identifies these families with the corresponding continuous versions: intersect on a countable dense parameter set and then use continuity. Thus the pin change also holds simultaneously throughout each compact positive parameter interval. For every deterministic segment, $H^{(10)}(x,y) = H^{(1)}(x,y) - U$, so

$$\xi^{-1}\log\rho_\xi^{(10)}(x,y) - H^{(10)}(x,y) = \xi^{-1}\log\rho_\xi^{(1)}(x,y) - H^{(1)}(x,y).$$

Thus the centered logarithmic remainder is exactly unchanged. Gaussian exponential moments of U preserve all fixed spatial moments used below. Until the final pin-restoration paragraph, ρ_ξ, h, H refer to the radius-ten pin. We do not transfer an exponentially small comparison failure by intersecting with a fixed-probability pin bound.

Proposition 3.1 and the similarity identity (3.15) give

$$\log\rho_\xi(x,y) = \alpha(\xi)\log r + \xi H(x,y) + \xi R_\xi^{x,y}, \qquad \left\|R_\xi^{x,y}\right\|_{L^p} \le C_p\xi^{1/24}, \tag{7.1}$$

where $r = |x-y|$ and $H(x,y) = \left\langle h, \mu_{[x,y]}\right\rangle$. This is the same $H(x,y)$ for every parameter.

Set $Z_\xi = \log(1+\rho_\xi)$, as a compact-profile-valued variable. Proposition 3.7 of [4] gives fixed B, M, C, c such that, for $0 < v < 1/8$,

$$\mathbb{P}[\|Z_\xi - Z_\eta\|_R > v] \le Cv^{-M}e^{-cv^M n} \tag{7.2}$$

for $\xi, \eta \ge a$ in a fixed small parameter interval, provided, with $\tau = |\xi - \eta|$,

$$\tau \le e^{-Ca^{-B}v^{-B}}, \qquad n \ge Cv^{-M}\log(C/v), \qquad \tau n \le cv. \tag{7.3}$$

Its first condition can be omitted if the pair already satisfies the fixed-confidence comparison

$$\mathbb{P}[\|\rho_\xi - \rho_\eta\|_{R_s} > c_s v] \le p_s \tag{7.4}$$

on the fixed guarded seed ball, with the prescribed constants c_s, p_s. This last clause is essential.

Let $\theta = 1/24$. For $\xi, \eta \in [a, 2a]$, (7.1) and Lipschitz continuity of α imply, uniformly over deterministic endpoints in each fixed ball $\overline{B}_R$,

$$\|\rho_\xi(x,y) - \rho_\eta(x,y)\|_{L^p} \le C_{R,p}(\tau + a^{1+\theta}). \tag{7.5}$$

For $r \le 1$, use $\|H(x,y)\|_{L^p} \le C_p\sqrt{1+\log(1/r)}$, the bound $\|\rho_\xi(x,y)\|_{L^p} \le C_p r^{1/2}$ at sufficiently small parameters, and $|e^s - e^t| \le |s-t|\,(e^s + e^t)$. The factor $r^{1/2}(1+|\log r|)$ is bounded. For $r \ge 1$ in a compact ball the same estimate is immediate.

Fix $0 < \beta < \theta_1 < \theta$. Spatial grids, the upper spatial Hölder moments in [2, Proposition 3.2], and arbitrarily high fixed moments in (7.5) show that

$$\left\|\|\rho_\xi - \rho_\eta\|_R\right\|_{L^p} \le C_{R,p}a^{1+\theta_1} \quad \text{if } \tau \le a^{1+\theta}. \tag{7.6}$$

Explicitly, choose a spatial exponent $\sigma < 1$, a mesh a^K with $K\sigma > 1+\theta_1$, and a fixed moment $q \ge p$ with $2dK/q < \theta - \theta_1$. The grid maximum and its interpolation have the asserted bound. At accuracy $v = a^{1+\beta}$, (7.6) therefore supplies the seed (7.4) for all sufficiently small a.

7.2 Finite grids and the full interval

The moment estimate controls a polynomial number of initial parameter values. We use the comparison estimate to fill the gaps with finer nested grids, distributing a summable error over successive levels. Fix $v = a^{1+\beta}$, and choose a fixed large D, for example

$$D > B(2+\beta) + B + M + 3.$$

For sufficiently large fixed A, let G_j be the relative dyadic grid on $[a, 2a]$ of level

$$k_j = \lceil Aa^{-D}2^{Dj} \rceil, \qquad w_j = a2^{-k_j}, \qquad v_j = 2^{-j-8}v.$$

Compare adjacent points of G_j and every point of G_{j+1} with both bracketing points of G_j. Use $n_j = c_0 v_j / w_j$ in (7.2), with a fixed small c_0. All conditions (7.3) hold. The total failure is at most

$$\sum_{j \geq 0} C 2^{k_{j+1}} v_j^{-M} \exp\{-c v_j^{M+1}/w_j\} = O_m(a^m) \tag{7.7}$$

for every fixed m. Indeed the positive terms in the logarithm are at most $Ca^{-D}2^{D(j+1)} + Cj + C\log(1/a)$; the negative term contains

$$c\, a^{(1+\beta)(M+1)-1} 2^{-(M+1)j} \exp\{cAa^{-D}2^{Dj}\},$$

which dominates their sum uniformly in j.

Choose $L_1 > D + (1+\beta)M + 3$ and a relative dyadic base grid whose spacing w_{base} is comparable to $c_0 v a^{L_1}$, rounded downward to a dyadic spacing. It is nested in G_0 for sufficiently small a. Its cardinality is $O(a^{-L_1-\beta})$, and $w_{\text{base}} \leq a^{1+\theta}$ for sufficiently small a. At each base point, (7.1) at the unit pair and exponential Taylor expansion give

$$\|\rho_\xi(0, e_1) - 1 - \xi H\|_{L^p} \leq C_p a^{1+\theta}, \qquad H = \left\langle h, \mu_{[0,e_1]} \right\rangle.$$

Markov's inequality controls every base value to accuracy cv, with failure

$$C_p a^{-L_1-\beta+p(\theta-\beta)}.$$

Choose the fixed p after L_1, D so that this is $O(a^m)$.

Compare each node of G_0 to both bracketing base points at tolerance cv and depth $n = a^{-L_1}$. The seed (7.4) is available, so the generic exponential spacing requirement is omitted. The sum of these failures is at most

$$Cv^{-M} \exp\{Ca^{-D} - cv^M a^{-L_1}\} = O_m(a^m). \tag{7.8}$$

The drift condition follows from $w_{\text{base}} n \leq cv$. The fine-grid errors sum to $O(v)$. On $|H| \leq a^{-1}$, whose complement has a Gaussian tail, moving the linear term to a bracketing base point costs at most $w_{\text{base}}\,|H| = O(v)$. On this event the base unit distances are uniformly bounded, so the transform $\log(1+\cdot)$ can be inverted with a common Lipschitz constant at all controlled unit values.

The continuous mean-log version constructed in [4, Section 5, proof of Theorem 1.1] extends the bound from the dense union of the G_j to the whole slab. Equations (7.7)–(7.8) give a summable failure bound on the slabs $a = 2^{-k}$, taking any fixed failure power $m > 1$. Borel–Cantelli yields

$$\rho_\xi(0, e_1) = 1 + \xi H + O_{h,\beta}(\xi^{1+\beta}), \qquad \beta < 1/24.$$

Intersect the events for a countable sequence of positive exponents increasing to $1/24$. The bound for any smaller exponent follows from the bound at one larger member of this sequence, so one event supports all the exponents in Proposition 7.1. Restore the unit-sphere pin before normalizing by a median. Since U is one almost surely finite random variable,

$$\rho_\xi^{(1)}(0,e_1) = e^{\xi U}\rho_\xi^{(10)}(0,e_1) = 1 + \xi(H^{(10)} + U) + O_{h,\beta}(\xi^{1+\beta}) = 1 + \xi H^{(1)} + O_{h,\beta}(\xi^{1+\beta}).$$

This completes the proof of Proposition 7.1. The change to point-median normalization is made for every dimension in the next section.

8 Normalization and completion

The first-order estimates so far use mean-log normalization. The scalar relating this normalization to the point median is deterministic. We control that scalar using the positive density of the limiting Gaussian, and then apply the same conversion to the almost sure and L^p estimates.

Lemma 8.1 (Stability of the median)**.** *Let H be a centered nondegenerate Gaussian and let $Z_\xi = H + R_\xi$. Suppose that for every fixed $1 \le q < \infty$, $\|R_\xi\|_{L^q} \le C_q\xi^\nu$ for all sufficiently small ξ. Then for every $0 < r < \nu$,*

$$|\operatorname{med} Z_\xi| \le C_r\xi^r \qquad \text{for all sufficiently small } \xi\,.$$

Proof. The distribution function of H has derivative bounded below by a positive constant near zero. Set $t = A\xi^{\nu q/(q+1)}$. Markov's inequality gives

$$\mathbb{P}[|R_\xi| > t] \le C_q A^{-q}\xi^{\nu q/(q+1)}\,.$$

Choose A large enough that this is smaller than half the increase of the Gaussian distribution function over an interval of length t near zero. Then

$$\mathbb{P}[Z_\xi \le -2t] \le \mathbb{P}[H \le -t] + \mathbb{P}[|R_\xi| > t] < \frac{1}{2}, \qquad \mathbb{P}[Z_\xi \le 2t] \ge \mathbb{P}[H \le t] - \mathbb{P}[|R_\xi| > t] > \frac{1}{2}\,.$$

The lower median lies in $[-2t, 2t]$. For the prescribed r, choose the fixed q so that $\nu q/(q+1) > r$. □

Apply this lemma to $Z_\xi = (\rho_\xi(0,e_1) - 1)/\xi$ and $H = H_d$, using (3.7). Since a positive affine map preserves the lower median,

$$m_\xi = 1 + O(\xi^{1+r}) \qquad \text{for every } 0 < r < 1/24\,. \tag{8.1}$$

In particular, m_ξ is bounded away from zero for all sufficiently small ξ. This estimate holds under the original unit-sphere pin in every dimension.

The exponent bounds also quantify the parameter conversion. From $Q_d(\xi) = d/\gamma + \gamma/2$ and $\xi Q_d(\xi) \le 1 + \sqrt{2}\xi$, we obtain $\xi \le (Q_d(\xi) - \sqrt{2})^{-1}$, so $\xi \to 0$ as $\gamma \to 0$. Equation (1.6) then gives $\xi \asymp \gamma$ and

$$\frac{\xi}{\gamma} = \frac{\alpha(\xi)}{d + \gamma^2/2} = \frac{1}{d} + O(\gamma)\,. \tag{8.2}$$

Proof of Theorem 1.1. Let $\widehat{D}_\xi = \rho_\xi/m_\xi$, and put $\delta_\xi = (m_\xi - 1)/\xi$. The exact identity

$$\frac{\widehat{D}_\xi(0,e_1) - 1}{\xi} - H_d = \frac{Z_\xi - H_d - \delta_\xi}{m_\xi} + \left(\frac{1}{m_\xi} - 1\right) H_d \tag{8.3}$$

makes the effect of normalization explicit. Fix $p < \infty$ and $0 < r < 1/24$. Equations (3.7) and (8.1), together with $H_d \in L^p$, give

$$\left\| \frac{\widehat{D}_\xi(0, e_1) - 1}{\xi} - H_d \right\|_{L^p} \le C_{d,p,r} \xi^r \,. \tag{8.4}$$

This proves the needed moment estimate directly, without deducing it from almost sure convergence.

For the pathwise estimate, use Proposition 6.1 at the unit pair when $d = 2$, and Proposition 7.1 when $d \ge 3$. They give, on the stated common event,

$$Z_\xi = H_d + O_{h,d,\beta}(\xi^\beta), \qquad 0 < \beta < \beta_d \,.$$

For each such β, choose r strictly between β and $1/24$. Insert (8.1) into (8.3). The resulting pathwise remainder is still $O_{h,d,\beta}(\xi^\beta)$. The events from the two parameter propositions already support every allowed exponent by a countable intersection; the median corrections are deterministic.

Finally, $D_h^\gamma = \widehat{D}_{\xi(\gamma)}$, and

$$\frac{D_h^\gamma(0, e_1) - 1}{\gamma} - \frac{1}{d} H_d = \frac{\xi}{\gamma} \left(\frac{\widehat{D}_\xi(0, e_1) - 1}{\xi} - H_d \right) + \left(\frac{\xi}{\gamma} - \frac{1}{d} \right) H_d \,. \tag{8.5}$$

Equation (8.2) proves both (1.9) and (1.10). The mean-log pathwise bounds hold for every sufficiently small real ξ, so their substitution at $\xi(\gamma)$ holds for every sufficiently small real γ. □

Proof of Theorem 1.2. Let $K \subset \mathbb{C}$ be compact and enclose it in an integer-radius ball. The continuum part of Proposition 5.1 gives

$$\left\| \sup_{x,y \in K} |\rho_\xi(x, y) - |x - y| - \xi \mathcal{L}_h(x, y)| \right\|_{L^p} \le C_{K,p} \xi^{1+1/512} \,.$$

Use (8.1) with a fixed $r \in (1/512, 1/24)$. Dividing by m_ξ changes the displayed residual by $O_{L^p}(\xi^{1+r})$, since the compact diameters and $\sup_{K^2} |\mathcal{L}_h|$ have every fixed moment. The same calculation with Proposition 6.1 gives the pathwise bound for every $\beta < 1/512$. After division by γ, the additional parameter error from (8.2) is $O_{L^p}(\gamma)$ and $O_h(\gamma)$ on each compact set. Integer balls and the common event in Proposition 6.1 give the stated simultaneous compact-set assertion. □

Corollary 8.2 (Opposite-face normalization)**.** *Let the metric instead be normalized by the lower median of its unrestricted distance between opposite faces of the unit cube, in the convention of [2, Section 7]. Then, for every fixed $d \ge 2$, its unit-pair first-order limit is*

$$\frac{1}{d}(H_d - q_d) \qquad \textit{almost surely and in every finite } L^p, \qquad q_d = \operatorname{med} \min_{y \in [0,1]^{d-1}} \left\langle h, \mu_{[(0,y),(1,y)]} \right\rangle < 0 \,.$$

Proof. In point-median units, the deterministic opposite-face median is $1 + \xi q_d + o(\xi)$ by [2, Corollary 7.1 and its proof]. Divide the unit-pair expansion by this deterministic scalar, and use (8.2). The scalar tends to one and its first-order correction tends to q_d, so the operation preserves both modes of convergence. □

A Audit record

Three successive complete drafts underwent model-assisted mathematical, source, and exposition review. Each round included a fresh reconstruction of the full argument and a separate check of its cited inputs. The revisions were checked again in the following round; the final correction was checked against the third reviewed draft.

First round. The rewrite made the Gaussian derivative coefficient, fixed normalization, and measurable geodesic selection explicit. It fixed the radii before coarse-field removal and placed the radial approximation and the geodesics used for rigidity on the same ball. The literal tolerance ranges of the comparison inputs were also restored.

Second round. The rewrite separated the deterministic similarity increment from the common Gaussian increment and the stochastic remainder. It distinguished the good-event threshold from the resulting metric-error threshold. The review also checked the order of moment and grid choices, both parameter constructions, and the simultaneous identification of the field pins.

Third round. A fresh complete reconstruction found no remaining mathematical obstruction under the stated inputs. The final rewrite made the proposed status of the higher-dimensional characterization explicit in the abstract and introduction. The checks covered the full real-parameter event, every fixed finite moment, and both deterministic normalization changes.

These reviews concern the deductions in this paper. The metric characterization in [1], the regularity estimates in [2], and the quantitative comparisons in [3, 4] remain the expressly stated inputs. In particular, the higher-dimensional theorem retains its conditional scope.

References

[1] *Higher-dimensional exponential metrics: A proposed axiomatic existence and uniqueness proof.* Supplied research manuscript, revised version in the user's archive, `higher-dimensional-lqg-metric-revised.pdf`. Theorem 1.2 and the covariance consequences in Section 10 are used as metric inputs.

[2] *First-order fluctuations of small-parameter exponential metrics.* Supplied research manuscript, `unified-first-order-exponential-metrics.pdf`, 16 pages. Included in the user's archive `LQG_Results_2026-10-17_to_19`. The uniform metric regularity is Proposition 3.2. The present paper reproduces the logarithmic first-order argument from these estimates. Corollary 7.1 is used for the optional opposite-face normalization.

[3] *The small-coupling Euclidean limit of the Liouville quantum gravity metric.* Supplied research manuscript, 19 October 2026, `lqg_almost_sure_euclidean_limit.pdf`, 47 pages. The inputs used here are Proposition 3.1, Theorem 4.1, Proposition 5.2, Theorem 10.1, Proposition 11.1, equation (11.13), and the simultaneous construction in Theorem 1.1.

[4] *Almost sure continuity of the higher-dimensional LQG metric.* Supplied research manuscript, `higher_dimensional_lqg_continuity.pdf`. Theorem 1.1 and Proposition 3.7 are used under that manuscript's strong-metric characterization input.

[5] J. Ding, E. Gwynne, and Z. Zhuang. *Tightness of exponential metrics for log-correlated Gaussian fields in arbitrary dimension.* arXiv:2310.03996v5, version 5, 2026.